

Arduino Based Robotic Kit is a versatile development kit featuring Arduino Nano, L293D motor driver, OLED display, and sensor sockets. Ideal for rapid prototyping of autonomous, Bluetooth-controlled, or IoT-enabled robotic systems, it provides an educational platform for students, researchers, and hobbyists to develop smart robot and embedded projects.

Key Features

- Built-in Arduino Nano (ATmega328P)
- L293D motor driver for DC/stepper motors
- 0.96" OLED display interface
- HC-05 Bluetooth module socket
- MQ-series gas sensor socket
- Onboard buzzer for alerts
- Multiple GPIO headers for sensors/servos
- Reverse-polarity & overcurrent protection



Fig: Robotic Kit

Technical Specification

Parameter	Value
Microcontroller	ATmega328P (Arduino Nano compatible)
Operating Voltage	5V DC; Input: 7–12V DC (Barrel/USB-C)
Motor Driver	L293D Dual H-Bridge, 2 channels (DC/stepper), 4.5–36V, 600mA per channel
Communication	HC-05 Bluetooth, I ² C, UART, SPI (optional)
Display	0.96" OLED (I ² C)
Sensors	MQ-series gas, ultrasonic, potentiometer
Buzzer	Onboard piezo
Power Supply	5V regulated, DC barrel jack, USB-C, onboard regulator
Protection	Reverse-polarity (SS34), overcurrent
Expansion	Female headers for I/O, GND, 5V, VIN, Reset
Optional Modules	Ultrasonic, IR, Servo, LCD

Applications

- Bluetooth-controlled robotic cars and vehicles
- Line follower and obstacle-avoiding robots
- Gas detection and alert systems (MQ-series sensors)
- IoT-based environmental monitoring and data logging
- Smart home and automation projects
- Educational robotics and embedded systems training
- Prototyping of embedded control systems and mobile robots
- Alarm and security alert systems
- Bluetooth-based appliance control and automation